

3.1 COILS

MIKROELEKTRONIKA

Coils are not a very common component in electronic circuits, however when they are used, they need to be understood. They are encountered in oscillators, radio-receivers, transmitter and similar devices containing oscillatory circuits. In amateur devices, coils can be made by winding one or more layers of insulated copper wire onto a former such as PVC, cardboard, etc. Factory-made coils come in different shapes and sizes, but the common feature for all is an insulated body with turns of copper wire.

The basic characteristic of every coil is its inductance. Inductance is measured in Henry (H), but more common are millihenry (mH) and microhenry (μH) as one Henry is quite a high inductance value. As a reminder:

$$1\text{H} = 1000\text{mH} = 10^6\mu\text{H}.$$

Coil inductance is marked by X_L , and can be calculated using the following formula:

$$X_L = 2\pi fL,$$

where f represents the frequency of the voltage in Hz and the L represents the coil inductance in H.

For example, if f equals 684 kHz, while $L=0.6$ mH, coil impedance will be:

$$X_L = 2 \cdot 3,14 \cdot 684000 \cdot 0,6 \cdot 10^{-3} = 2\,577\,\Omega.$$

The same coil would have three times higher impedance at three times higher frequency. As can be seen from the formula above, coil impedance is in direct proportion to frequency, so that coils, as well as capacitors, are used in circuits for filtering at specified frequencies. Note that coil impedance equals zero for DC ($f=0$).

Several coils are shown on the figures 3.1, 3.2, 3.3, and 3.4.

The simplest coil is a single-layer air core coil. It is made on a cylindrical insulator (PVC, cardboard, etc.), as shown in figure 3.1. In the figure 3.1a, turns have space left between them, while the common practice is to wind the wire with no space between turns. To prevent the coil unwinding, the ends should be put through small holes as shown in the figure.

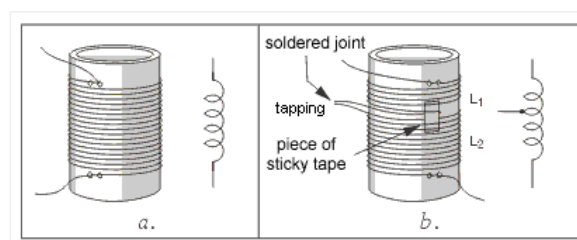


Fig. 3.1: Single-layer coil

Figure 3.1b shows how the coil is made. If the coil needs 120 turns with a tapping on the thirtieth turn, there are

two coils L1 with 30 turns and L2 with 90 turns. When the end of the first and the beginning of the second coil are soldered, we get a “tapping.”

A multilayered coil is shown in figure 3.2a. The inside of the plastic former has a screw-thread, so that the ferromagnetic core in the shape of a small screw can be inserted. Screwing the core moves it along the axis and into the center of the coil to increase the inductance. In this manner, fine changes to the inductance can be made.

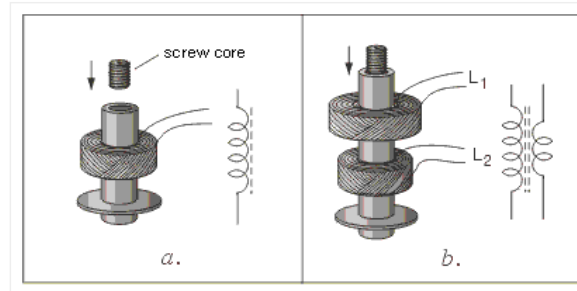


Fig. 3.2: a. Multi-layered coil with core, b. Coupled coils

Figure 3.2b shows a high-frequency transformer. As can be seen, these are two coils are coupled by magnetic induction on a shared body. When the coils are required to have exact inductance values, each coil has a ferromagnetic core that can be adjusted along the coil axis.

At very high frequencies (above 50MHz) coil inductance is small, so coils need only a few turns. These coils are made of thick copper wire (approx. 0.5mm) with no coil body, as shown on the figure 3.3a. Their inductance can be adjusted by physically stretching or squeezing the turns together.

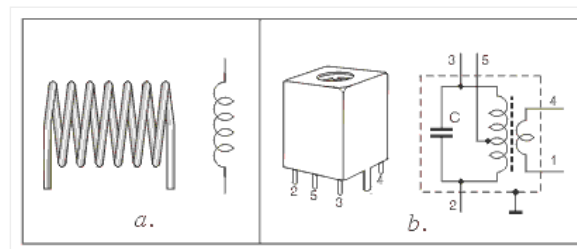


Fig. 3.3: a. High frequency coil, b. Inter-frequency transformer

Figure 3.3b shows a metal casing containing two coils, with the schematic on the right. The parallel connection of the first coil and capacitor C forms an oscillatory circuit. The second coil is used for transferring the signal to the next stage. This is used in radio-receivers and similar devices. The metal casing serves as a screen to prevent external signals affecting the coils. For the casing to be effective, it must be earthed.

Fig 3.4 shows a “pot core” inductor. The core is made in two halves and are glued together. The core is made of ferromagnetic material, commonly called “ferrite.” These inductors are used at frequencies up to 100kHz. Adjustment of the inductance can be made by the brass or steel screw in the centre of the coil.

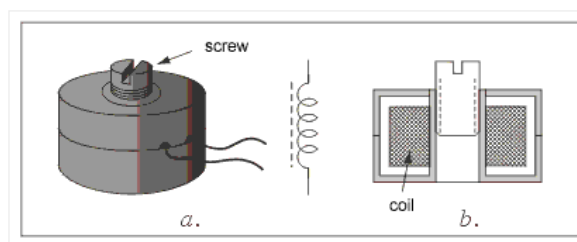


Fig. 3.4: A “pot core” inductor



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